A

PROJECT REPORT

ON

“Echohub”

SUBMITTED AS A PARTIAL REQUIREMENT
FOR THE DEGREE OF
BACHELOR OF COMPUTER APPLICATION
(B.C.A)



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SHREE UTTAR GUJARAT BCA COLLEGE AFFILIATED TO VEER NARMAD SOUTH GUJARAT UNIVERSITY, SURAT YEAR 2024-25



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Date: 03/10/2023

Certificate

This is to certify that the Mini Project Report entitled ***"Echohub"*** *have been submitted by* ***Mr. Sahil Kanojiya (Exam No:4463)*** *towards partial fulfillment of the requirement for 5th semester of Bachelor Of Computer Application* ***(B.C.A.)*** *for the academic year* ***2024-25***.

Signature	***Signature***	***Signature***
Project Guide	*External Examiner*	***(Prof. Pradip H. Rajput)***

ABSTRACT

- **Introduction**:
  - Echohub is a modern, interactive social media platform designed to user interaction.
  - The platform allows users to share content, connect with others .
- **Objectives**:
  - To provide a seamless and engaging social networking experience.
  - To facilitate content creation, and communication among diverse user .
- **Key Features**:
  - User-generated content in the form of posts, images, and videos.
  - Interactive features like likes, comments, shares, and save.
  - Customizable user profiles with privacy controls.
- **User Management**:
  - Multi-tier user roles: admins, regular users.
  - Role-based permissions to manage access to content and platform features.
- **Security & Privacy**:
  - Strong data encryption and two-factor authentication for user protection.
  - Tools for reporting, flagging, and moderating inappropriate content.
- **Conclusion**:
  - Echohub aims to be a dynamic and user-friendly social platform that encourages community interaction while maintaining user experience.

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1. INTRODUCTION

1.1 Project Description

1. **Purpose**:
    - Echohub is a social media platform designed to connect people and facilitate content sharing.
2. **Target Users**:
    - Individuals looking to engage with communities, share personal content, and interact online.
3. **Core Functionality**:
    - Users can create, share, and interact with posts, images, and videos.
4. **Engagement Features**:
    - Includes likes, comments, shares, and real-time messaging.
5. **Community Building**:
    - Offers tools for creating groups, events, and discussions.
6. **User Experience**:
    - Aims to provide a seamless, intuitive interface for easy navigation and interaction.
7. **Security**:
    - Focuses on user privacy and data protection through robust security measures.

1.2 Project Profile

Project Title	**Echohub**
Front End	**ReactJs, Tailwind Css, RTK**
Back End	**SpringBoot**
Database	**MongoDB**
Team Members	**2 Members**
Project Duration	**Approx 1.5 month- 2 month**
Type Of Application	**Web-Application**
Platform	**vs code,Intelij idea**
Operating System	**Windows 11**
Internal Guide	**Prof . Pooja Soni**
Submitted By	**Mr Sahil Kanojiya**
Submitted To	**Shree Uttar Gujrat BBA & BCA Collage**

2. ENVIRONMENT DESCRIPTION

2.1 Hardware And Software Requirements

Hardware and Software configuration for the implementation environment is as follows.

➢ Software Requirements

Operating System	**Microsoft Windows 11**
Frontend	**ReactJs**
Backend	**Java**
Scripting Language	**Javascript**
DataBase	**MongoDB**

➢ Hardware Requirements

Processor	**Any but Latest Is Good For Performance**
RAM	**8 GB**
Hard Disk	**259 GB**
Disk Space	**10 GB Minimum**

2.2 Technologies Used

Frontend	**Html,css,Bootstrap,Reactjs**
Backend	**Java (Spring Boot)**
Scripting Language	**Javascript**
DataBase	**MnogoDB**

3. SYSTEM ANALYSIS

3.1 Existing System And It's Drawbacks

- **Existing System**

  - **Basic Functionality**:
    - **User Accounts**: Allows users to register, log in, and manage profiles.
    - **Content Sharing**: Users can post text, images, and videos.
    - **Interactions**: Features include likes, comments.
    - **User Connectivity:** User can follow another users.

- **Drawbacks**

  - **Security Concerns**:
    - **Data Vulnerabilities**: Potential weaknesses in data encryption or authentication processes could expose user information to unauthorized access.
    - **Privacy Issues**: Inadequate privacy settings may lead to users' data being exposed or misused.

3.2 Expected Advantages

- **Robust Security and Privacy:**
  - **Data Protection**: Strong encryption and secure authentication methods to safeguard user data and privacy.
  - **Privacy Controls**: Comprehensive privacy settings allowing users to manage their personal information and control who can view their content.
  - manual intervention, saving both time and operational costs while providing a more reliable service experience.

4. PROPOSED SYSTEM

4.1 Scope

The scope of the system is defined on the basis of Various Functionalities provided by the system the scope can be explained as :-

- **Developing a web-based system**: The proposed system will be a fully functional webapplication that supports various user roles and functionalities.
- **User Information Management**: The system will handle all user-related information, including the creation of new user lists, retrieval, updates, anddeletion of user data.
- **Reduction of Manual Processes**: The system aims to replace manual processeswith automated solutions, thereby enhancing efficiency and accuracy.

4.2 Project Modules

The Echohub project is divided into several core modules that work together to create a fully functional social media platform. Each module has its own responsibilities and functions within the system.

1. User Management Module

- **Account Creation and Authentication**: Users can sign up, log in, and manage their profiles.
- **Role Management**: Different access levels for users, admins, and moderators.
- **User Profile Settings**: Users can update their personal details and account preferences.

2. Content Management Module

- **Content Posting**: Users can post text, images, and videos.
- **Editing and Deletion**: Users can edit or delete their content.
- **Engagement Features**: Likes, comments, shares, and reactions on posts.

3. Messaging and Notification Module

- **Push Notifications**: Alerts for new messages, likes, comments, and other interactions.

4. Feed and Discovery Module

- **Personalized Feed**: Displays content based on user preferences, interests, and interactions.
- **Content Discovery**: Tools for searching, exploring trending topics, and discovering new users and groups.

5. Analytics and Reporting Module

- **User Analytics**: Tracks user activity, engagement, and content performance.
- **Platform Metrics**: Helps admins monitor system performance and user trends.

6. Admin

- **Content Moderation**: Admins and moderators can monitor and manage content to ensure community guidelines are followed.
- **User Management**: Admins can handle user reports, ban users, or moderate content.
- **Platform Management**: Tools for admins to control system settings, updates, and platform-wide announcements.

4.3 Objectives/Functionalities

1. User Management

- Allow users to create accounts, log in, and manage their profiles.
- Support different roles (users, admins, moderators).

2. Sharing

- Users can post and share text, images, and videos.
- Enable interaction through likes, comments, and sharing.

3. Personalized Feed

- Deliver a customized feed based on user preferences and interactions.
- Help users discover relevant content and users.

4. Security and Privacy

- Without username and password not access the other person.

5. Monetization

- Offer ad management tools and targeted advertisements.
- Provide premium features through subscription models.

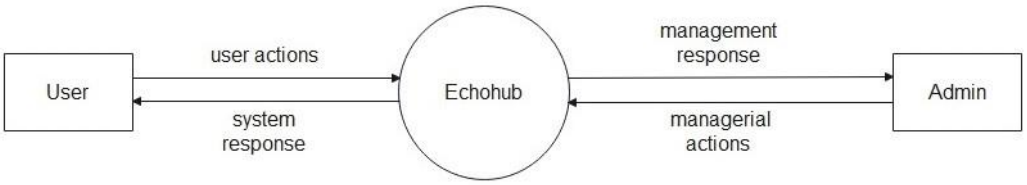
8. Analytics

- Track user activity, engagement, and content performance.
- Offer insights for advertisers and content creators.

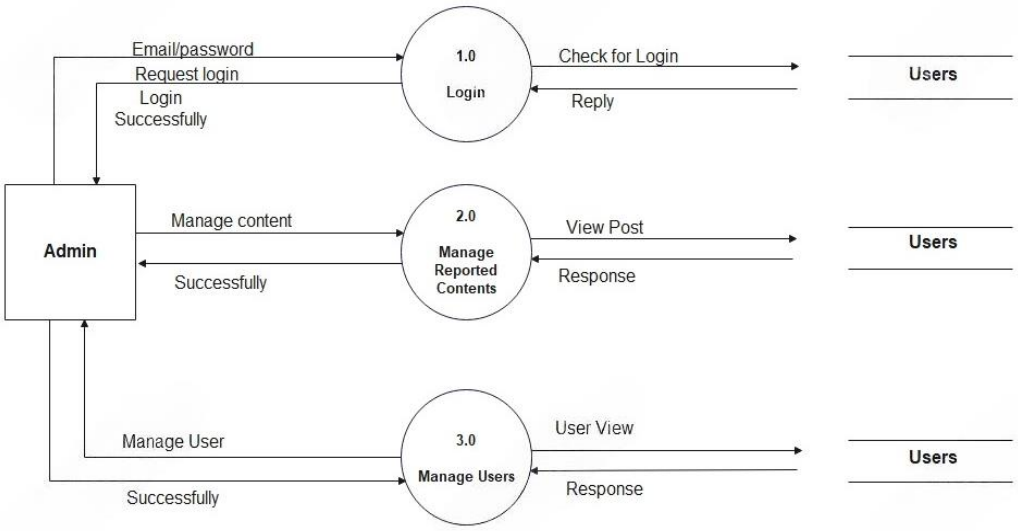
5.DETAIL PLANNING

5.1 Data Flow Diagram

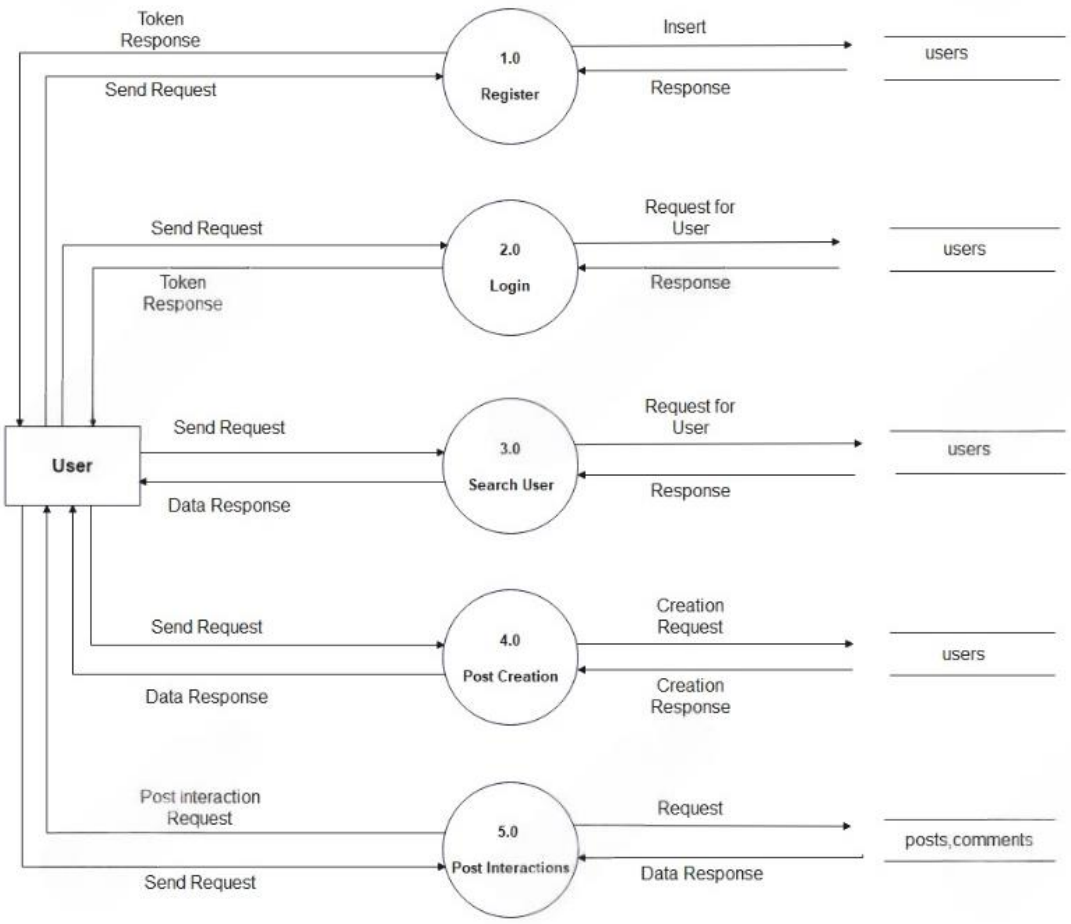
- **0 level – Context level**



➢ 1 level - Admin



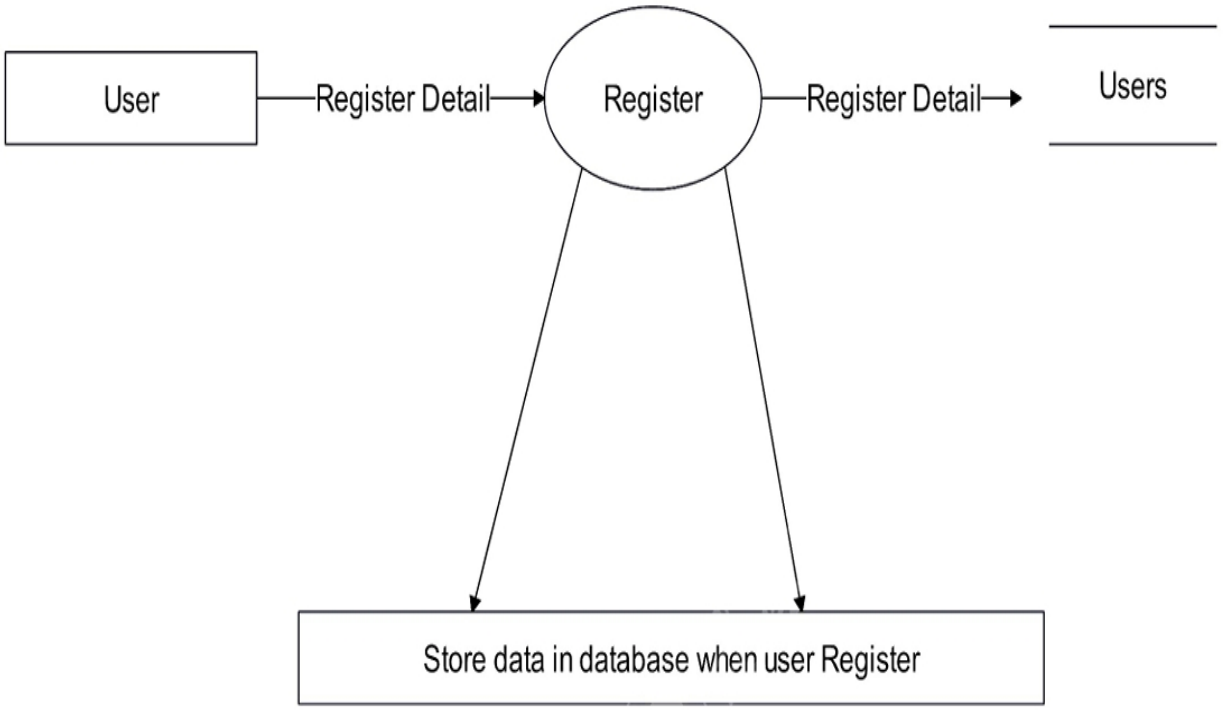
➢ level - User


Token
Response
Send Request
1.0
Register
Insert
Response
users
Send Request
Token
Response
2.0
Login
Request for
User
Response
users
User
Send Request
Data Response
3.0
Search User
Request for
User
Response
users
Send Request
Data Response
4.0
Post Creation
Creation
Request
Creation
Response
users
Post interaction
Request
Send Request
5.0
Post Interactions
Request
Data Response
posts,comments

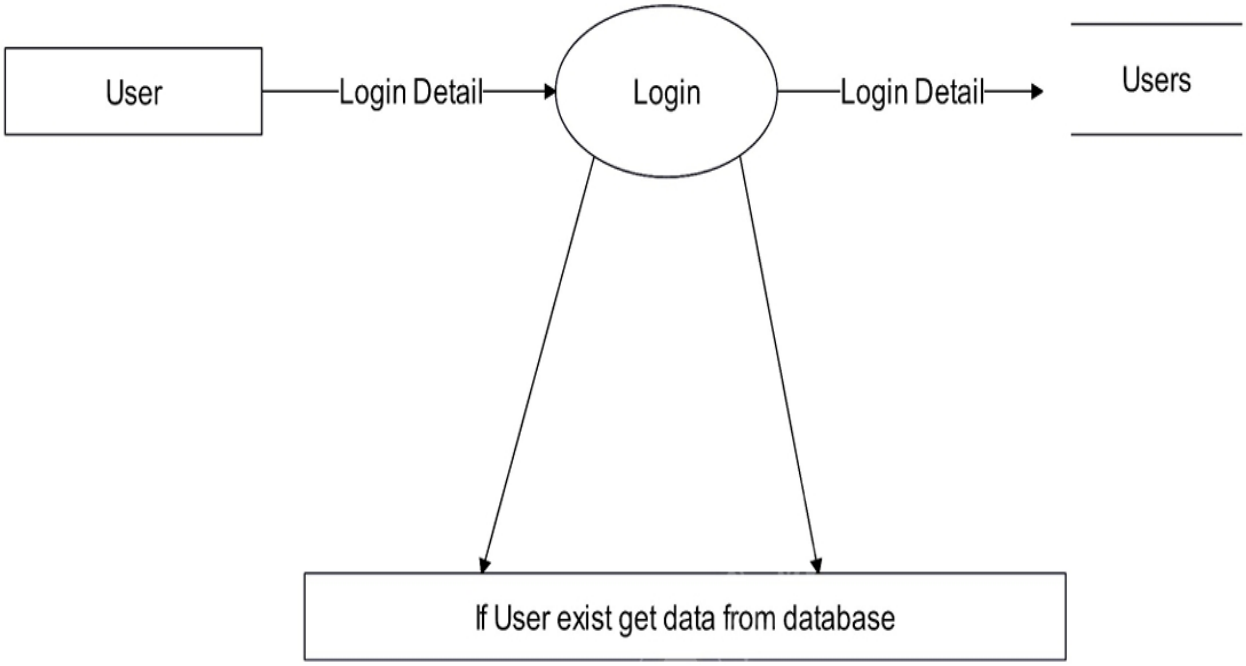
5.2 Process Specification

- A process specification is a method used to document , Analyze and explain the decision-making logic and formulas used to create output making logic and formulas used to create output data from process input data.

  - **Register process**



- **Login process**



5.3 Data Dictionary

1. Table Name : Users

Column Name	Data Type	Description	Default Value
_id	ObjectId	Unique identifier for the user	auto
Username	String	Username or email of the user	null
Fullname	String	Full name of the user	null
Email	String	Email address of the user	null
Password	String	Encrypted password (hashed using bcrypt)	null
profile_picture_url	String	URL to the user's profile picture	null
created_at	Date	Account creation timestamp	Current Timestamp
modified_at	Date	Last modification timestamp	Current Timestamp
refresh_token	Array	List of references to refresh tokens for the user	null
Followers	Array	List of users who follow this user	[] (empty array)
followings	Array	List of users that this user follows	[] (empty array)
Posts	Array	List of post IDs associated with the user	[] (empty array)
saved_posts	Array	List of post IDs saved by the user	[] (empty array)
Roles	Array	Array of roles assigned to the user (e.g., ROLE_USER)	["ROLE_USER"]

2. Table Name : Posts

Column Name	Data Type	Description	Default Value
_id	ObjectId	Unique identifier for the post.	auto
user.id	ObjectId	Unique identifier of the user.	null
content	String	The text content of the post.	""
image_url	String	URL of the image associated with the post.	""
Video_url	String	URL of video associated with the post.	“”
created_at	Date	Timestamp indicating when the post was created.	Current Timestamp
modified_at	Date	Timestamp indicating when the post was last modified.	Current Timestamp
likes	Array	List of users who liked the post (default is an empty array).	[]
comments	Array	List of comments on the post (default is an empty array).	[]
reported_by_users	Array	List of users who reported the post (default is an empty array).	[]
saved_by_users	Array	List of user IDs who saved the post (default is an empty array).	[]

3. Table Name: Reports

Column Name	Data Type	Description	Default Value
_id	ObjectId	Unique identifier for the report	auto
post_id	String	ID of the associated post	null
reported_by	String	ID of the user who reported the post	null
reason	String	Reason for reporting (e.g., HARASSMENT_BULLYING)	""
created_at	Date	Timestamp when the report was created	Current Timestamp

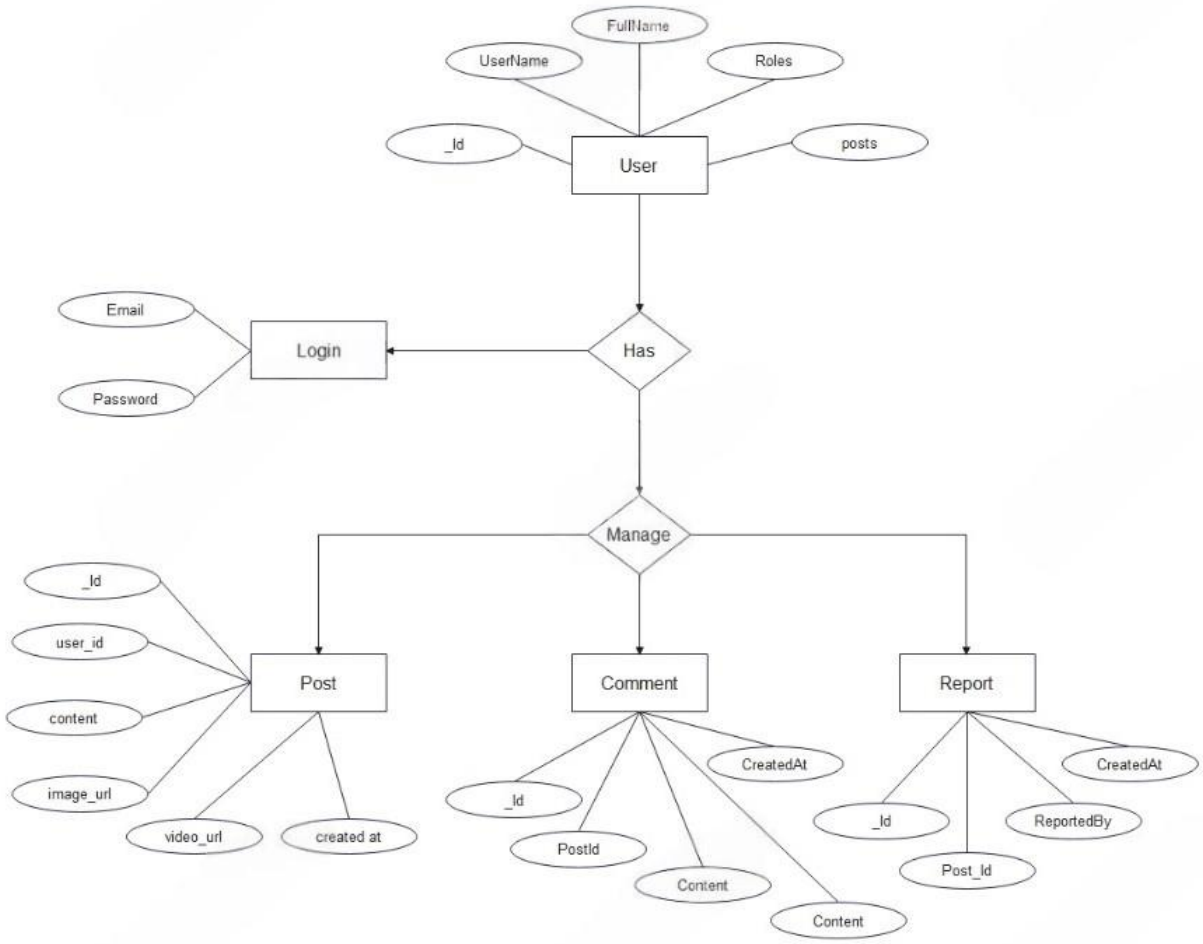
4. Table Name : Comment

Column Name	Data Type	Description	Default Value
_id	ObjectId	Unique comment identifier	auto
post_id	String	Associated post ID	“”
user	Reference	Creator of the comment	null
content	String	Text of the comment	""
created_at	Date	Comment creation timestamp	Current Timestamp
likes	Array	Users who liked the comment	[]

5. Table Name : Refresh_Tokens

Column Name	Data Type	Description	Default Value
_id	ObjectId	Unique identifier for the refresh token	auto
token	String	The refresh token string	“”
expiry_date	Date	Expiry date and time of the refresh token	Current Timestamp
username	String	Username associated with the refresh token	“”

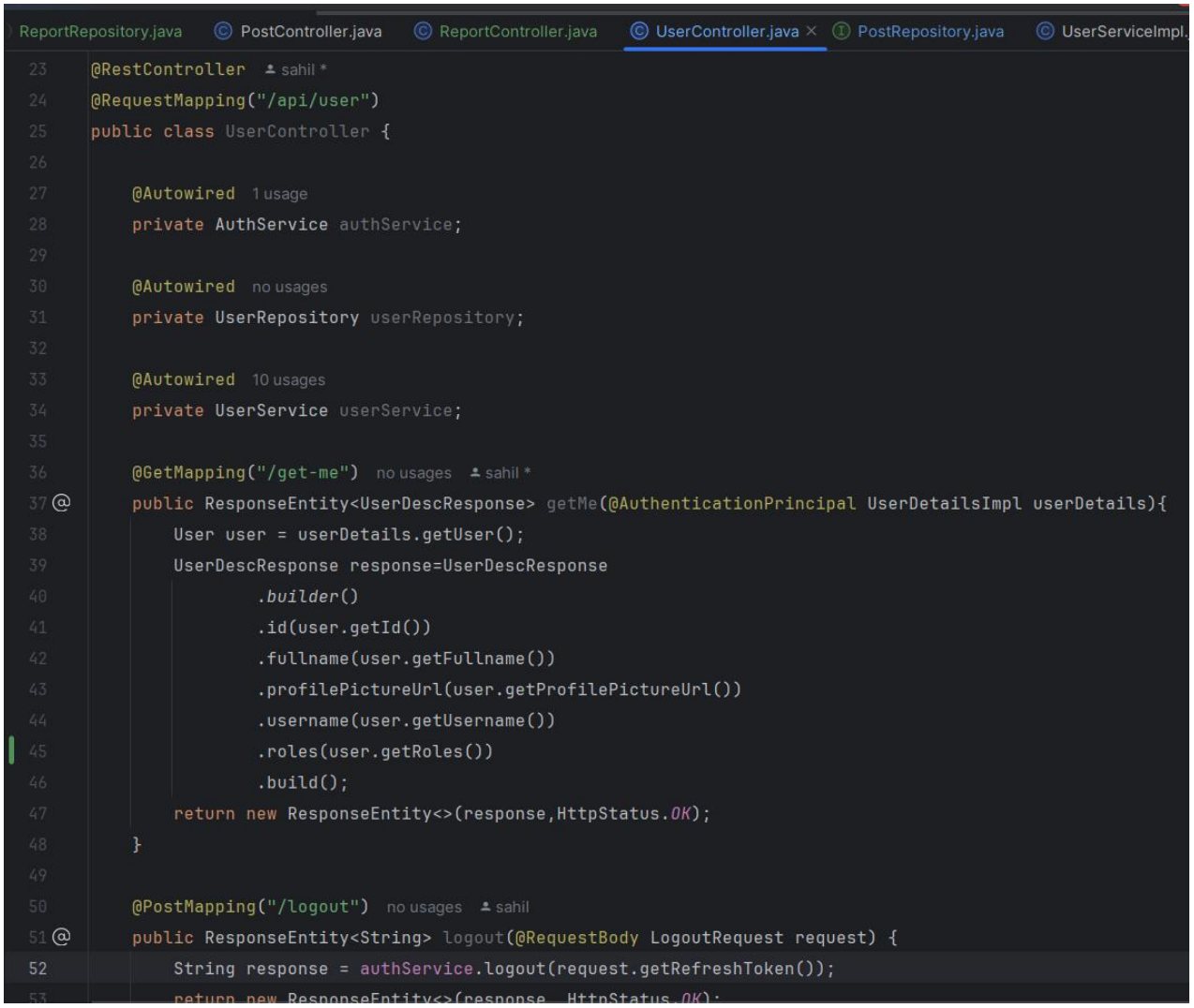
5.4 Entity-Relationship Diagram



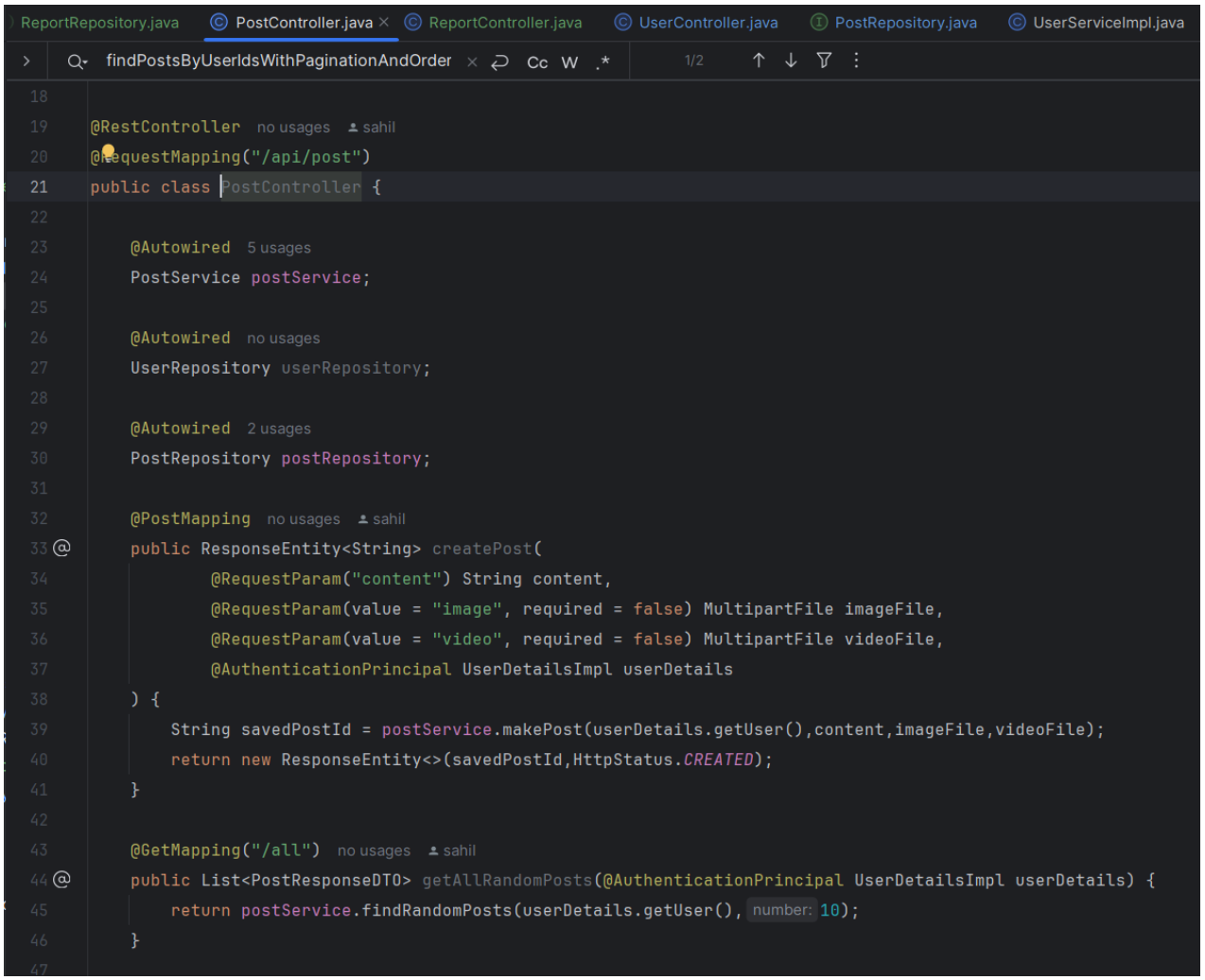
6. SYSTEM DESIGN

6.1 Input Design

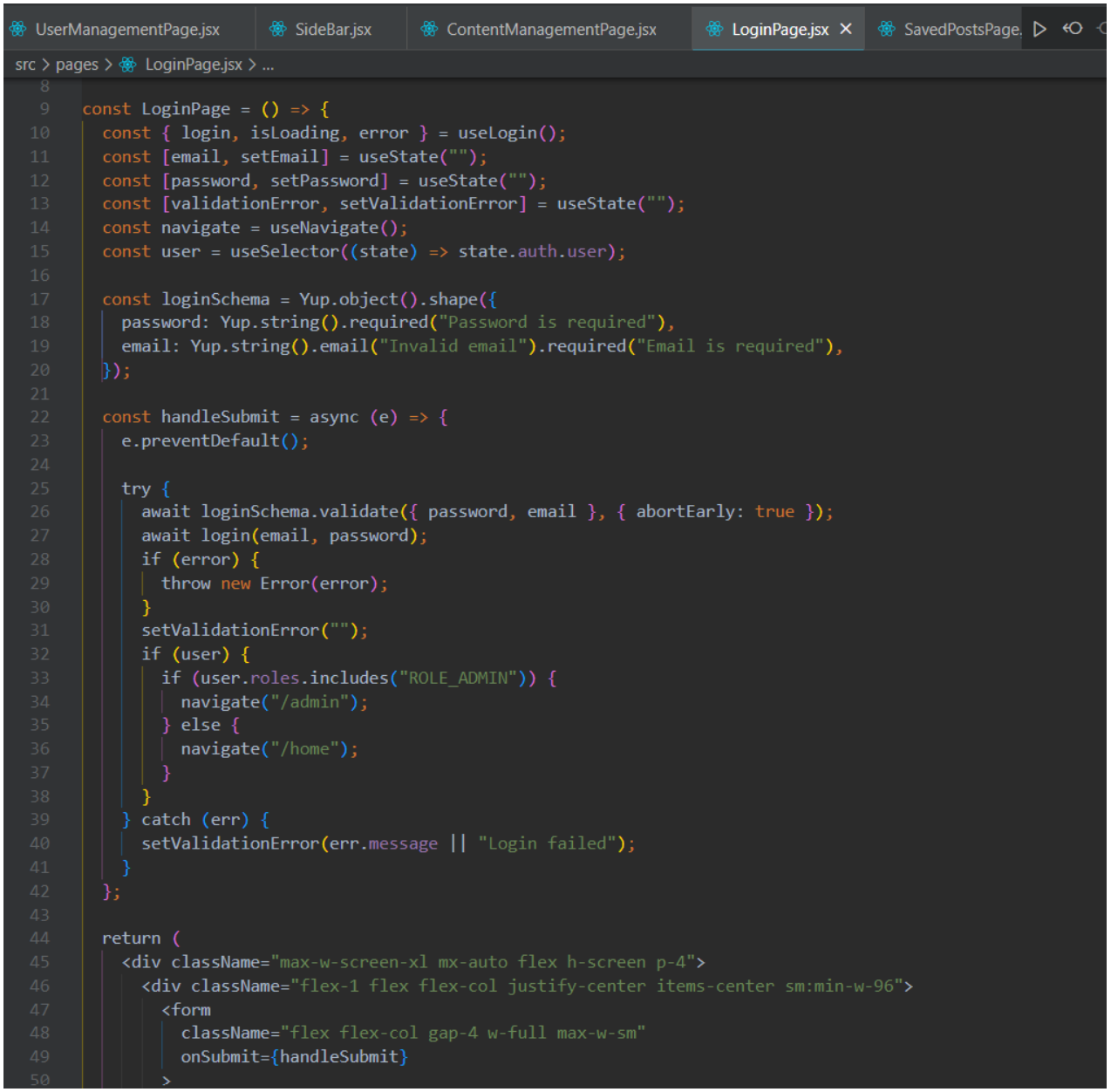
- **User Controller**



➢ Post Controller

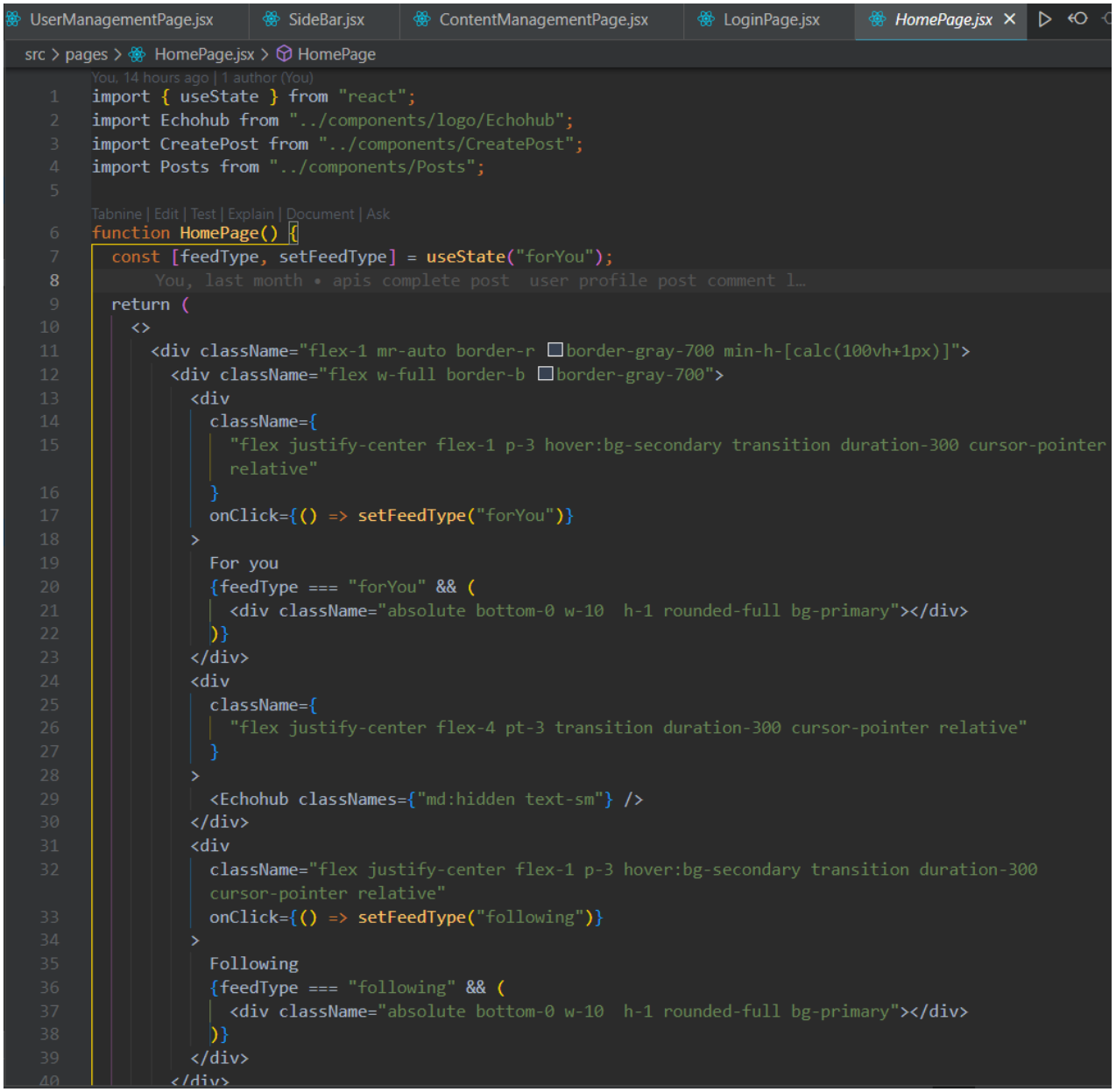


➢ Login Page


UserManagementPage.jsx
SideBar.jsx
ContentManagementPage.jsx
LoginPage.jsx
SavedPostsPage.
src > pages > LoginPage.jsx > ...
const LoginPage = () => {
const { login, isLoading, error } = useLogin();
const [email, setEmail] = useState("");
const [password, setPassword] = useState("");
const [validationError, setValidationError] = useState("");
const navigate = useNavigate();
const user = useSelector((state) => state.auth.user);
const loginSchema = Yup.object().shape({
password: Yup.string().required("Password is required"),
email: Yup.string().email("Invalid email").required("Email is required"),
});
const handleSubmit = async (e) => {
e.preventDefault();
try {
await loginSchema.validate({ password, email }, { abortEarly: true });
await login(email, password);
if (error) {
throw new Error(error);
}
setValidationError("");
if (user) {
if (user.roles.includes("ROLE_ADMIN")) {
navigate("/admin");
} else {
navigate("/home");
}
}
} catch (err) {
setValidationError(err.message || "Login failed");
}
};
return (

<form
className="flex flex-col gap-4 w-full max-w-sm"
onSubmit={handleSubmit}
>

➢ Home Page


UserManagementPage.jsx
SideBar.jsx
ContentManagementPage.jsx
LoginPage.jsx
HomePage.jsx
src > pages > HomePage.jsx > HomePage
You, 14 hours ago | 1 author (You)
import { useState } from "react";
import Echohub from "../components/logo/Echohub";
import CreatePost from "../components/CreatePost";
import Posts from "../components/Posts";
Tabnine | Edit | Test | Explain | Document | Ask
function HomePage() {
const [feedType, setFeedType] = useState("forYou");
You, last month • apis complete post user profile post comment l…
return (
<>

<div
className={
"flex justify-center flex-1 p-3 hover:bg-secondary transition duration-300 cursor-pointer relative"
}
onClick={() => setFeedType("forYou")}
>
For you
{feedType === "forYou" && (

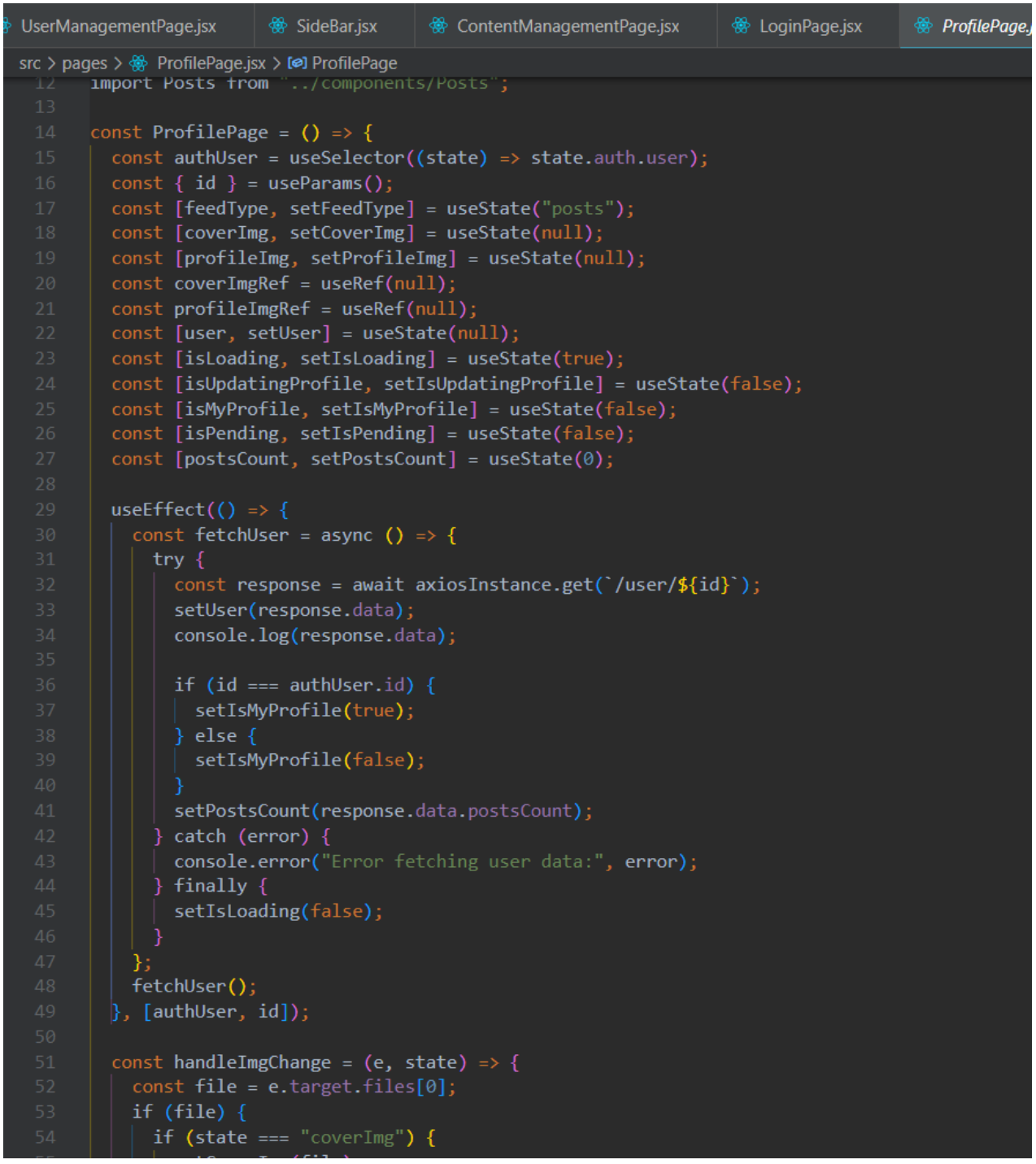
)}

<div
className={
"flex justify-center flex-4 pt-3 transition duration-300 cursor-pointer relative"
}
>
<Echohub classNames={"md:hidden text-sm"} />

<div
className="flex justify-center flex-1 p-3 hover:bg-secondary transition duration-300 cursor-pointer relative"
onClick={() => setFeedType("following")}
>
Following
{feedType === "following" && (

)}

➢ Profile Page




```jsx
import Posts from "../components/Posts";

const ProfilePage = () => {
  const authUser = useSelector((state) => state.auth.user);
  const { id } = useParams();
  const [feedType, setFeedType] = useState("posts");
  const [coverImg, setCoverImg] = useState(null);
  const [profileImg, setProfileImg] = useState(null);
  const coverImgRef = useRef(null);
  const profileImgRef = useRef(null);
  const [user, setUser] = useState(null);
  const [isLoading, setIsLoading] = useState(true);
  const [isUpdatingProfile, setIsUpdatingProfile] = useState(false);
  const [isMyProfile, setIsMyProfile] = useState(false);
  const [isPending, setIsPending] = useState(false);
  const [postsCount, setPostsCount] = useState(0);

  useEffect(() => {
    const fetchUser = async () => {
      try {
        const response = await axiosInstance.get(`/user/${id}`);
        setUser(response.data);
        console.log(response.data);

        if (id === authUser.id) {
          setIsMyProfile(true);
        } else {
          setIsMyProfile(false);
        }
        setPostsCount(response.data.postsCount);
      } catch (error) {
        console.error("Error fetching user data:", error);
      } finally {
        setIsLoading(false);
      }
    };
    fetchUser();
  }, [authUser, id]);

  const handleImgChange = (e, state) => {
    const file = e.target.files[0];
    if (file) {
      if (state === "coverImg") {
```

➢ Admin Page

UserManagementPage.jsx | SideBar.jsx | ContentManagementPage.jsx | LoginPage.jsx | AdminPage.jsx | SavedPosts

src > pages > AdminPage.jsx > AdminPage

```jsx
Tabnine | Edit | Test | Explain | Document | Ask
function AdminPage() {
  const [usersCount, setUsersCount] = useState(0);
  const [postsCount, setPostsCount] = useState(0);

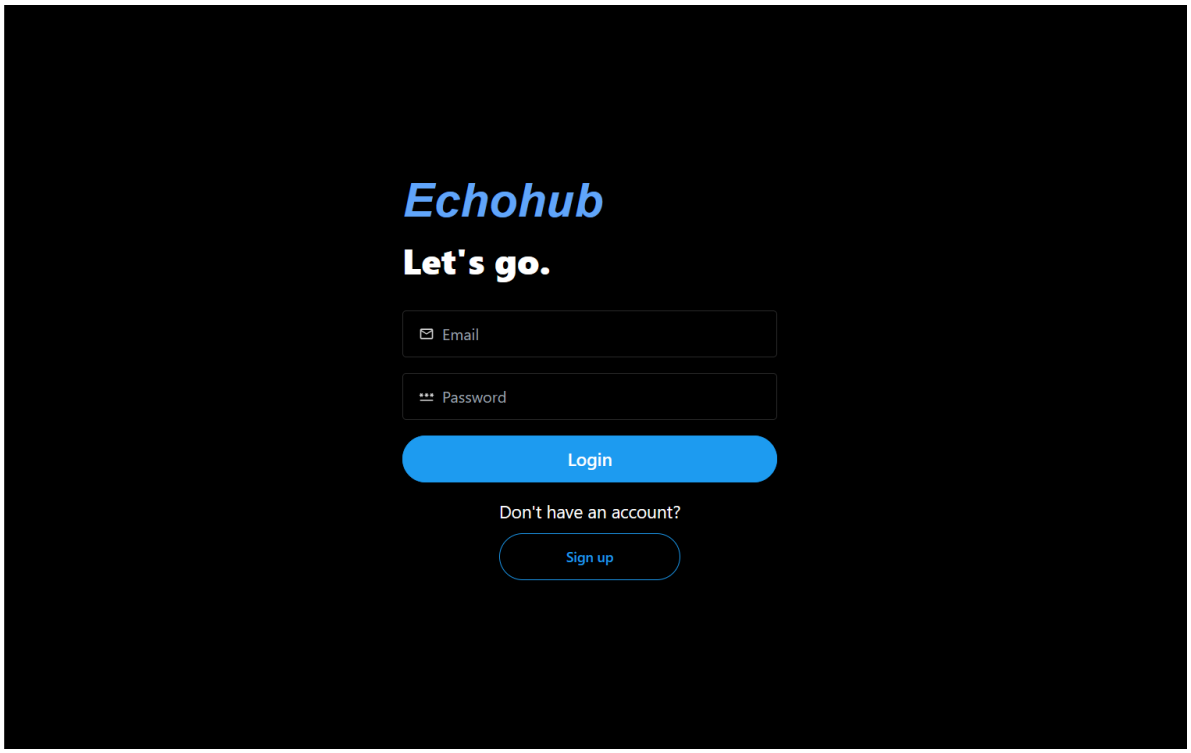
  useEffect(() => {
    const fetchDashboardData = async () => {
      try {
        const usersResponse = await axiosInstance.get("/admin/users/count");
        const postsResponse = await axiosInstance.get("/admin/posts/count");
        setUsersCount(usersResponse.data);
        setPostsCount(postsResponse.data);
      } catch (error) {
        console.error("Error fetching admin data", error);
      }
    };
    fetchDashboardData();
  }, []);

  const shutDown = async () => {
    const accessToken = getAccessToken();
    try {
      const response = await axios.post(
        "http://localhost:8080/actuator/shutdown",
        {},
        {
          headers: {
            Authorization: `Bearer ${accessToken}`,
          },
        }
      );
      console.log("Response from shutdown:", response.data);
      console.log("Shutting down...");
    } catch (error) {
      console.error("Error shutting down:", error);
      if (error.response) {
        console.error("Response data:", error.response.data);
      }
    }
  };

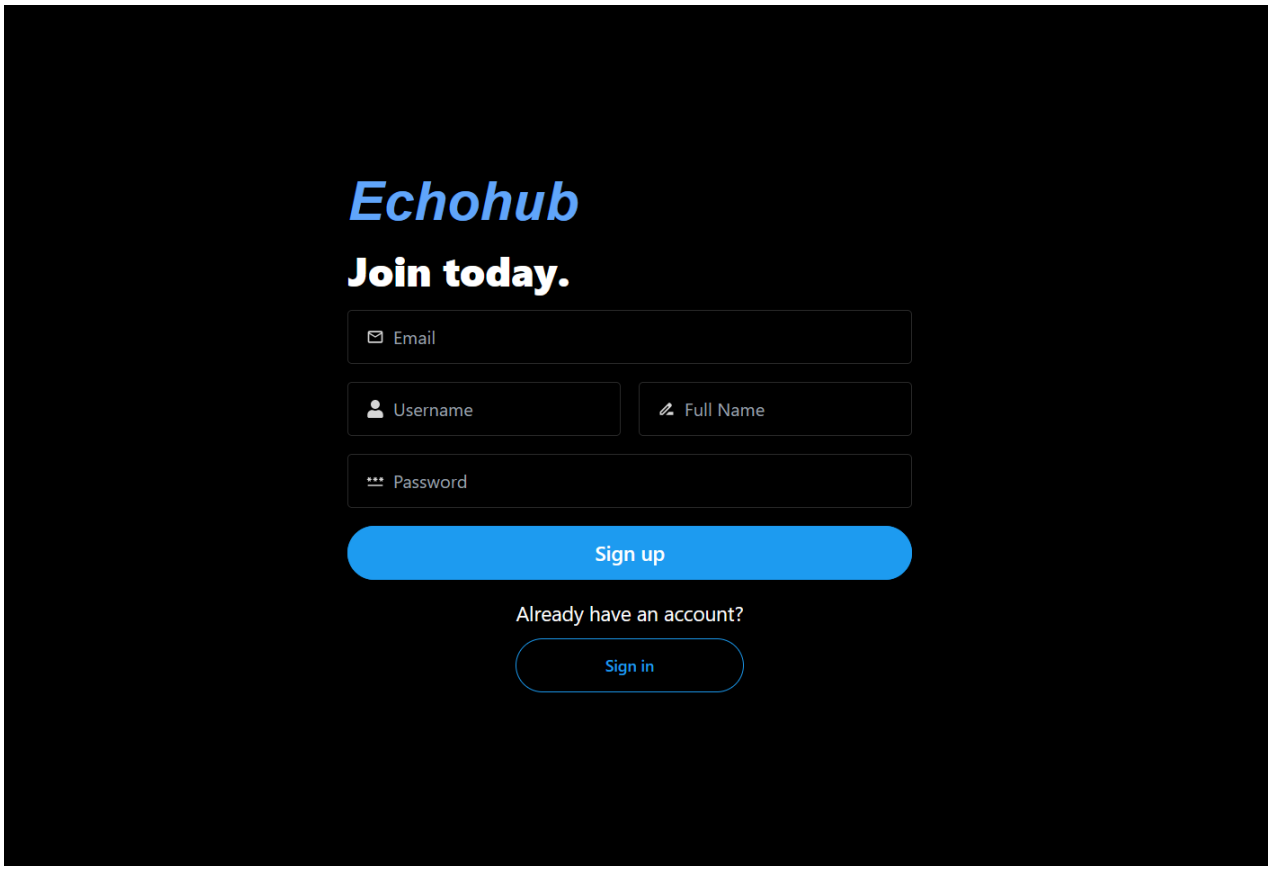
```

6.2 Output Design

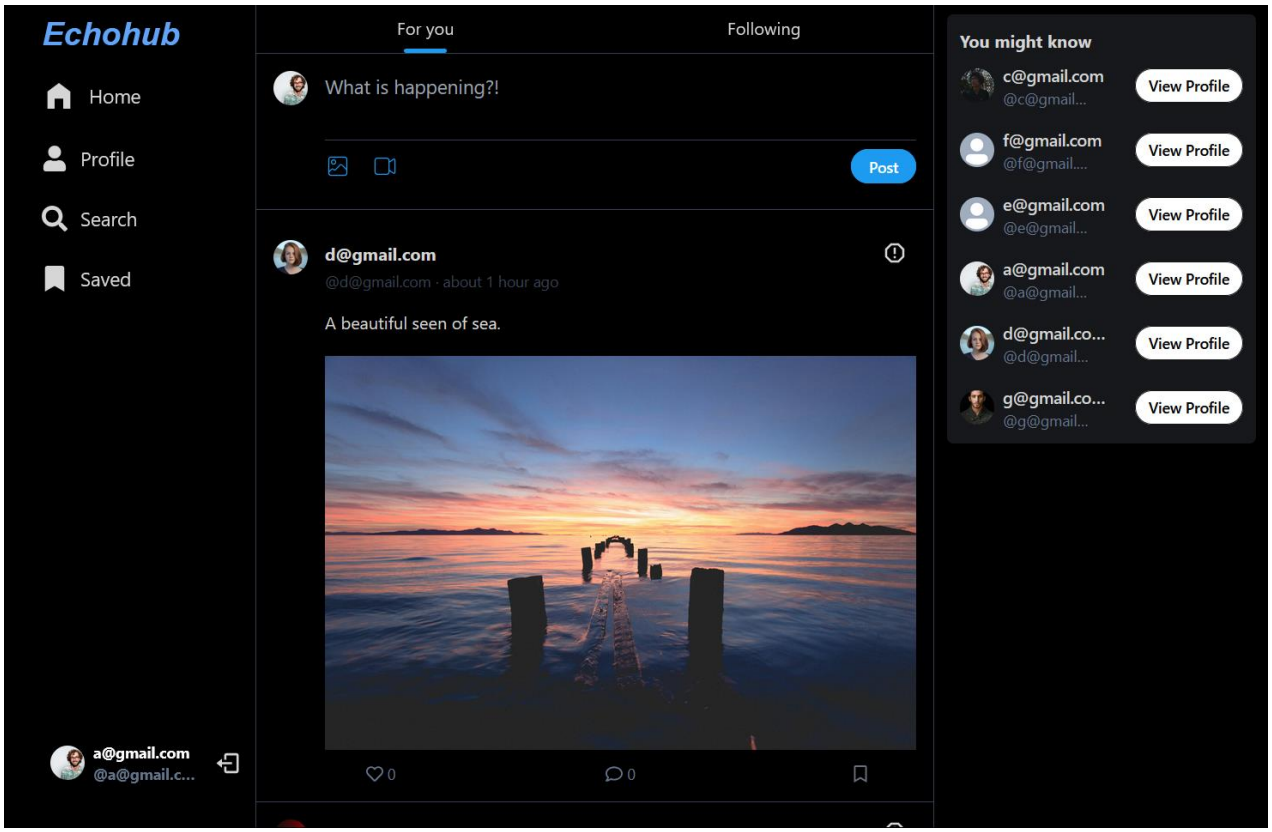
➢ Login Page



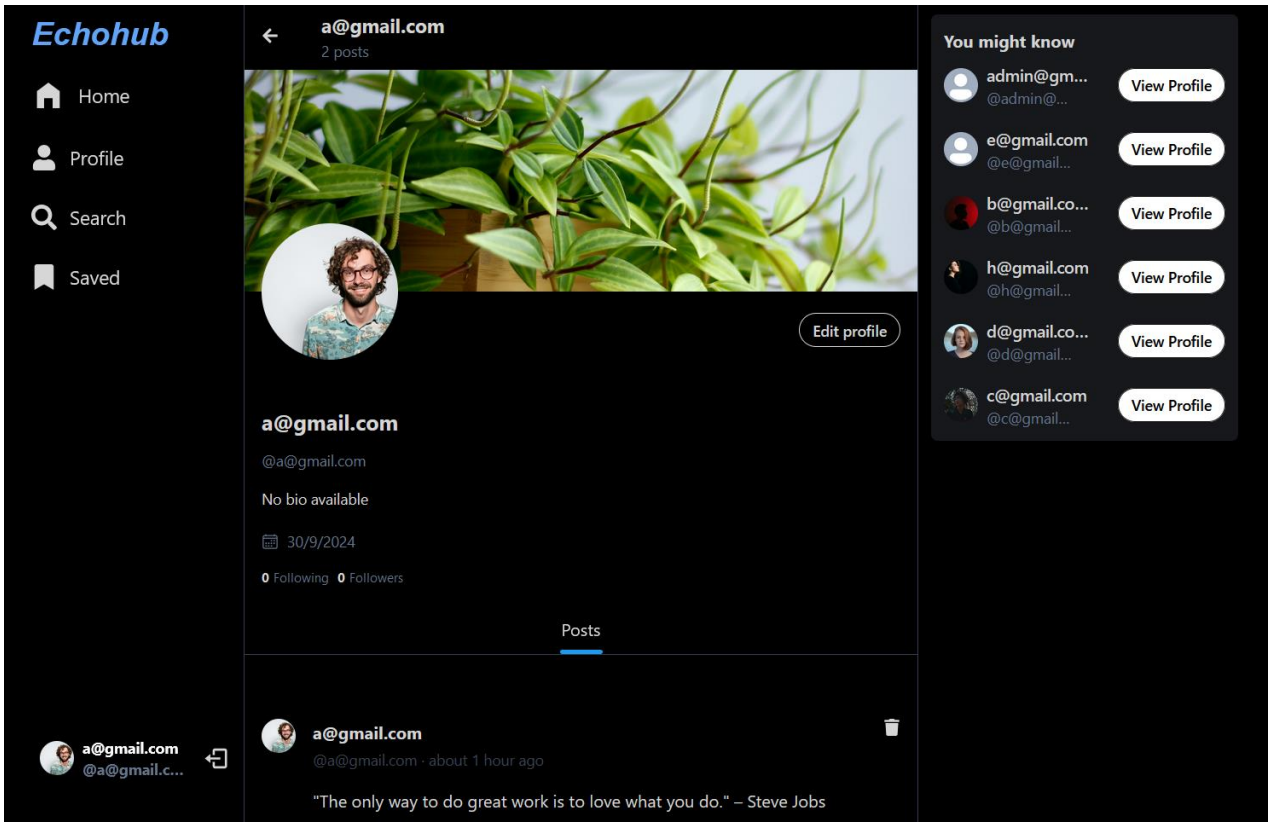
➢ Register Page



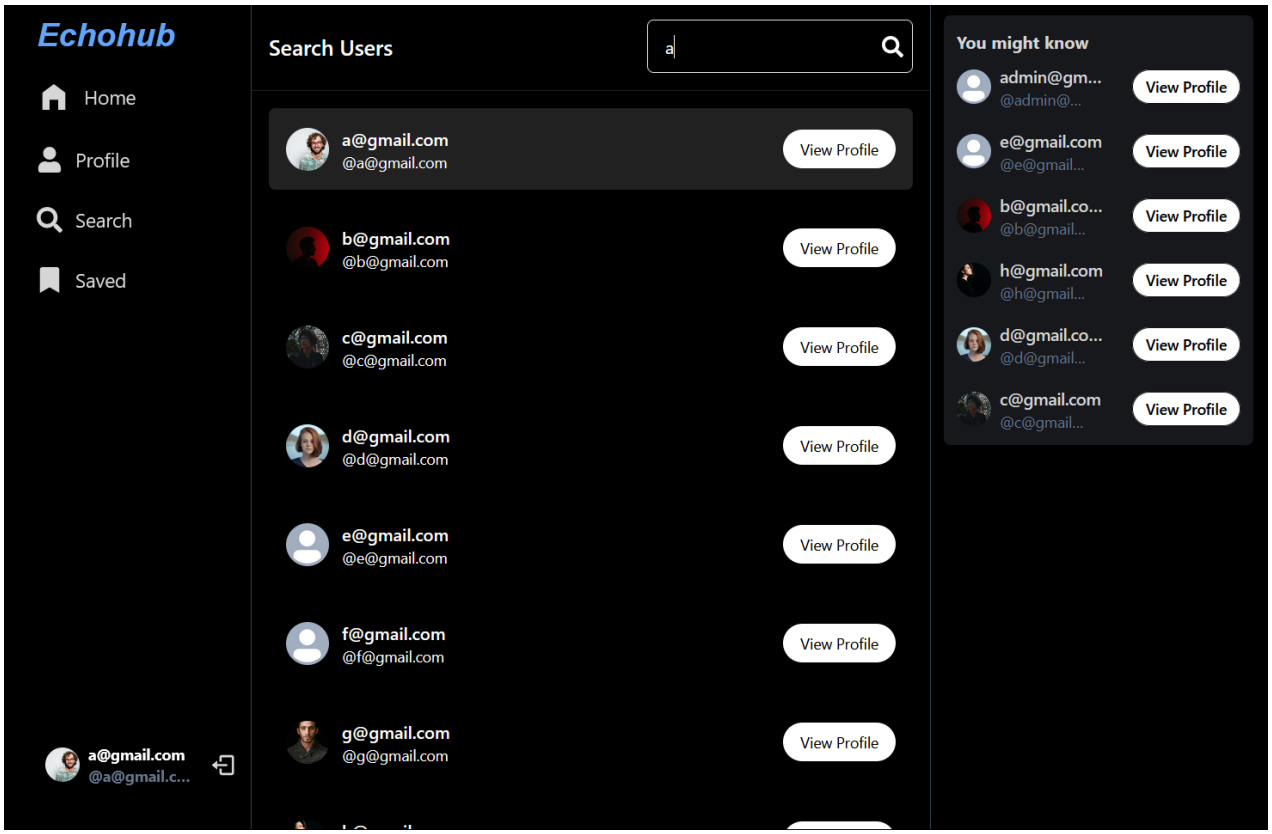
➢ Home Page


Echohub
Home
Profile
Search
Saved
a@gmail.com
@a@gmail.c...
For you
Following
What is happening?!
Post
d@gmail.com
@d@gmail.com · about 1 hour ago
A beautiful seen of sea.
0
0
You might know
c@gmail.com
@c@gmail...
View Profile
f@gmail.com
@f@gmail....
View Profile
e@gmail.com
@e@gmail...
View Profile
a@gmail.com
@a@gmail...
View Profile
d@gmail.co...
@d@gmail...
View Profile
g@gmail.co...
@g@gmail...
View Profile

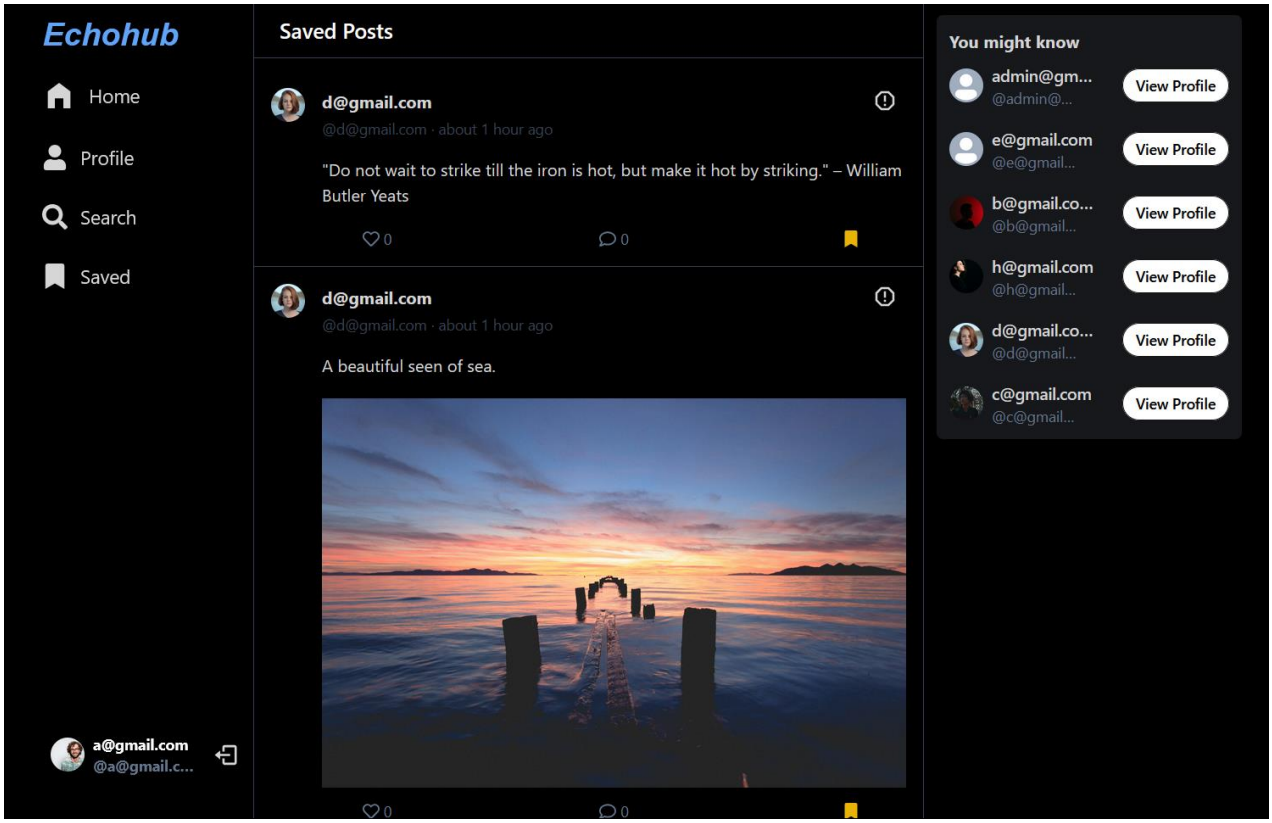
➢ Profile Page



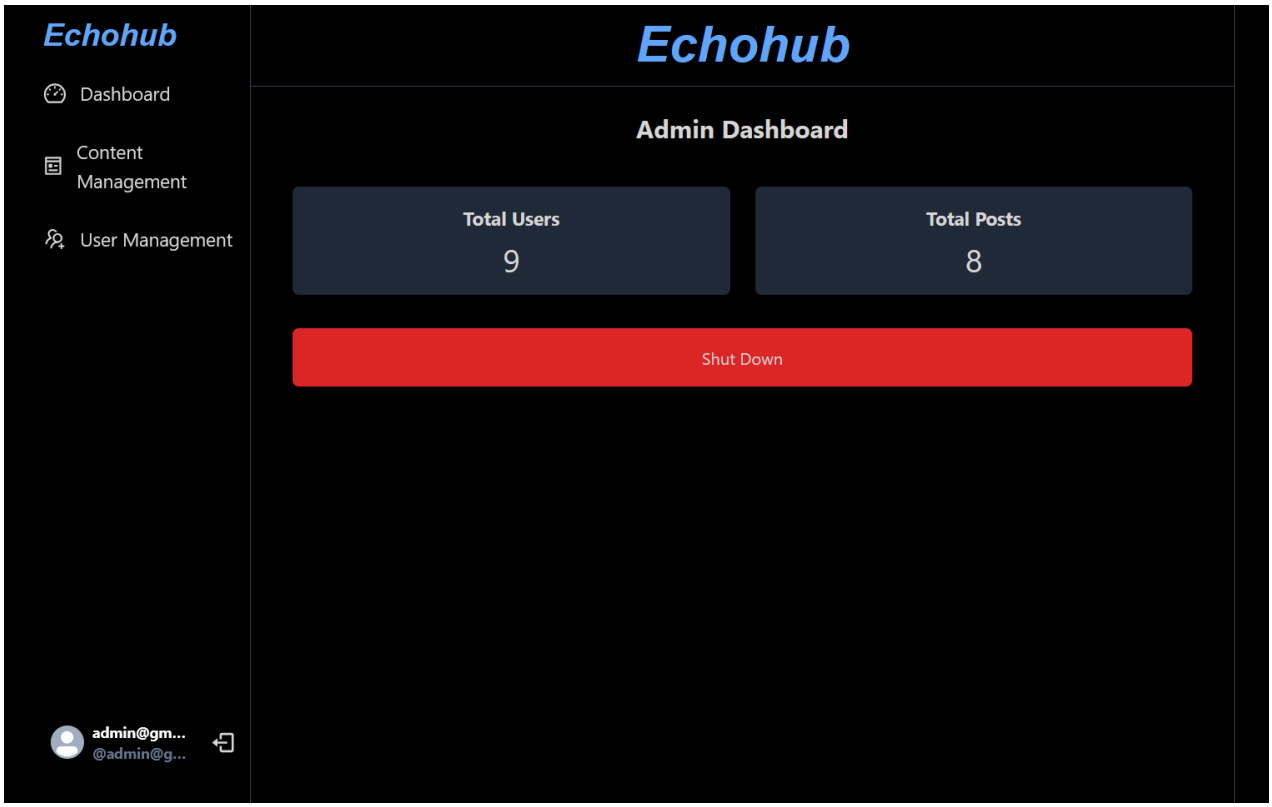
➢ Search Page


Echohub
Home
Profile
Search
Saved
a@gmail.com
@a@gmail.c...
Search Users
a
a@gmail.com
@a@gmail.com
View Profile
b@gmail.com
@b@gmail.com
View Profile
c@gmail.com
@c@gmail.com
View Profile
d@gmail.com
@d@gmail.com
View Profile
e@gmail.com
@e@gmail.com
View Profile
f@gmail.com
@f@gmail.com
View Profile
g@gmail.com
@g@gmail.com
View Profile
You might know
admin@gm...
@admin@...
View Profile
e@gmail.com
@e@gmail...
View Profile
b@gmail.co...
@b@gmail...
View Profile
h@gmail.com
@h@gmail...
View Profile
d@gmail.co...
@d@gmail...
View Profile
c@gmail.com
@c@gmail...
View Profile

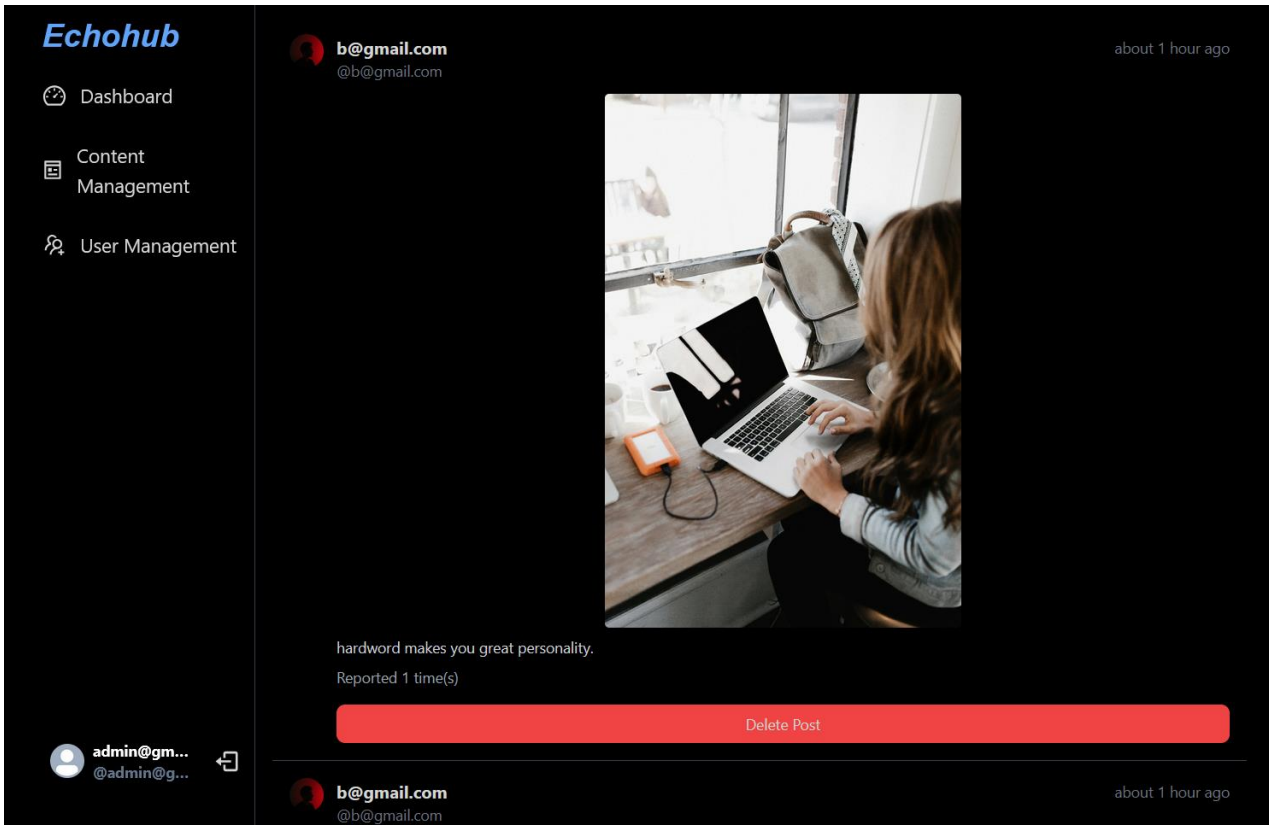
➢ Saved Post Page



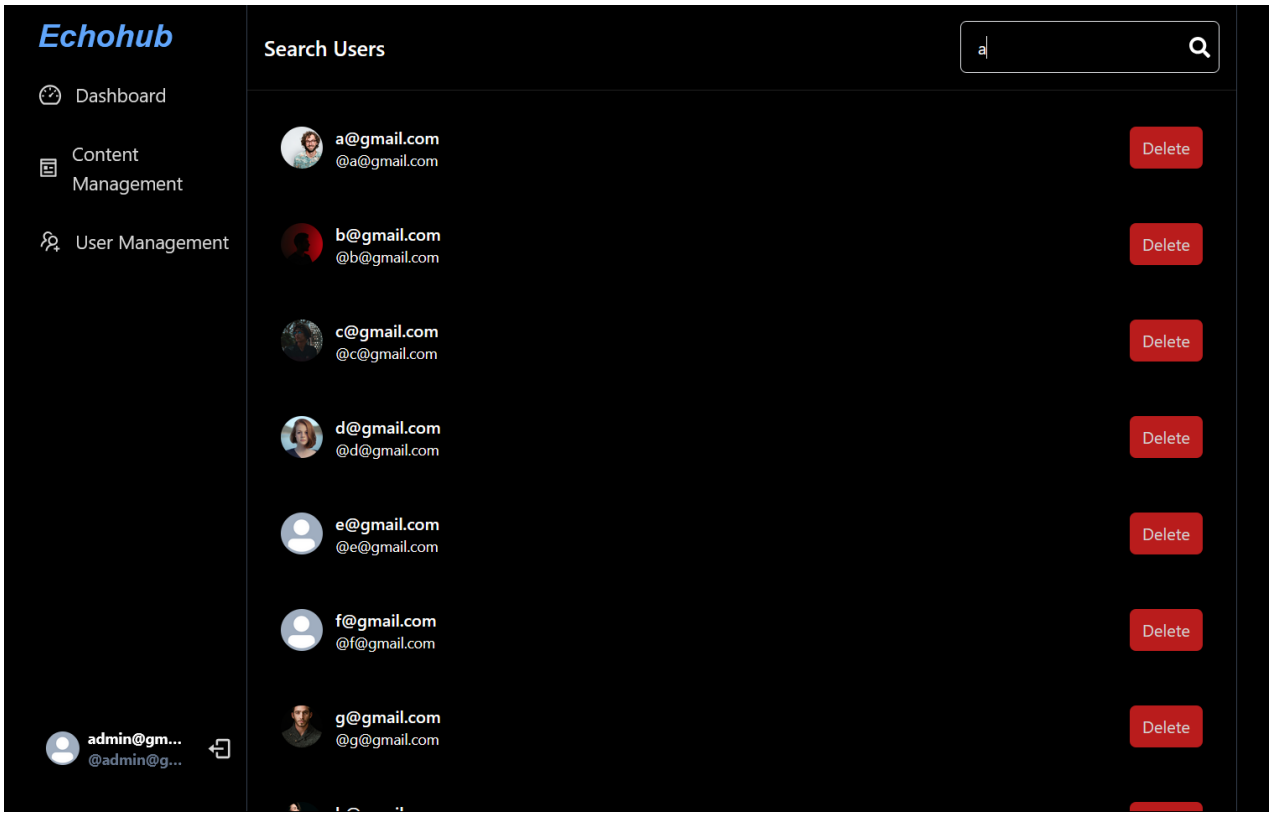
➢ Admin Dashboard



➢ Content Management Page



➢ User Management Page



7. LIMITATIONS AND FUTURE

6.1 Limitations

1. Content Moderation and Misinformation:

- Despite moderation efforts, Echohub might struggle with the spread of fake news, misinformation, and harmful content. Ensuring real-time monitoring and accuracy of posts can be challenging, leading to potential misinformation reaching a wide audience.

2. Echo Chambers and Polarization:

- The platform's follow-based system may create echo chambers, where users primarily engage with like-minded individuals or groups, limiting exposure to diverse perspectives and promoting polarization of opinions.

3. Online Harassment and Trolling:

- Public posts and anonymity features may lead to toxic behavior like trolling, harassment, or cyberbullying, creating a negative experience for some users.

4. Overload of Information:

- The fast-paced, real-time nature of Echohub can lead to an overwhelming amount of content, making it difficult for users to filter out noise and stay focused on relevant updates or conversations.

6.2 Future Scope of Enhancements:

1) **Advanced Content Filtering**:

    - Implement AI-driven filters to help users manage content overload by curating posts based on relevance, interests, and engagement.

2) **Improved Content Moderation**:

    - Introduce more advanced AI-based moderation tools to detect and prevent misinformation, hate speech, and harmful content in real-time.

3) **Expanded Post Formats**:

    - Allow for more diverse content types, such as polls, interactive stories, and live streams, enhancing user engagement beyond simple text and multimedia posts.

4) **Monetization Opportunities**:

    - Provide monetization options for creators and influencers through paid subscriptions, sponsored content, or direct support (e.g., tips, donations).

5) **Localization and Multilingual Support**:

    - Offer enhanced localization features, including language translations and region-specific trends, to make Echohub more accessible to a global audience.

6) **Collaborative Posting**:

    - Introduce features for group posts or collaborative threads, where multiple users can contribute to a single conversation or project.

7) **Video-based and Image based Posts**:

    - Add options for users to create short or video clips as posts, expanding communication styles beyond text.

8. REFERENCES

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